# GIRINATH M


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## PROFESSIONAL SUMMARY

Electronics and Communication Engineering student passionate about Embedded Systems, PCB Design, IoT and Electronics Hardware. Hands-on experience with ESP32, EasyEDA, Arduino and sensor-based projects through internships and academic work.

## EDUCATION

**Bachelor of Degree - Electronics and Communication Engineering.** **CGPA:** 8.5% **2024-2028**

NPR College of Engineering and Technology, Natham, Dindigul.

**HSC-**M.S.P. Solai Nadar Memorial Higher Secondary School, Dindigul. **Percentage:** 68.83% **2022-2024**

**SSLC-** Kalaimagal Higher Secondary School,Vadamadurai, Dindigul. **Percentage:** 64.4% **2017-2022**

## EXPERIENCE

**EMBEDDED & IoT Intern | Z00L00P** **20 Apr 2026 - Present**

- Designed and validated PCB schematics and layouts using EasyEDA in accordance with industry design standards.
- Evaluated electronic component datasheets to support accurate component selection and hardware design decisions.Conducted product research and contributed to Research & Development (R&D) activities for embedded and IoT-based solutions.

**IOT INTERN | TARCIN ROBOTIC LLP** **15 Dec 2025 - 07 Jan 2026**

- Developed an IoT-based radar detection system for real-time object detection and monitoring.
- Integrated sensors, microcontroller and wireless communication for accurate data transmission.
- Performed hardware integration, testing and system optimization to improve reliability and performance.

## TECHNICAL SKILLS

**Hardware:** Arduino , Esp32 **| PCB Design :** EasyEDA , KiCad , Proteus **| Tools :** Thingspeak , Github , VS Code **|Programming Language :** C , C++ , Python (Basics ) **| Certification:** NIELIT - VLSI for Beginners , Introduction for MATLAB & Simulink , Embedded for Beginners **| Language:** Tamil , English

## PROJECTS

### AIR QUALITY MONITORING SYSTEM — FEB'25

Developed an Arduino Uno-based air quality monitoring system using MQ-2, MQ-9, MQ-135 gas sensors and DHT11 sensor. Integrated an I2C LCD display for real-time monitoring of environmental parameters. Performed sensor interfacing, hardware integration and system testing.

### Radar Detection System — DEC'25

Developed an Arduino Uno-based radar detection system using an ultrasonic sensor and servo motor for real-time object detection.Designed a web interface to visualize detection data and improve user interaction. Integrated hardware components, sensor interfacing and embedded programming for accurate operation.

### IoT-Based Crop Root Health Monitoring System — APR'26

Developed an ESP32-based IoT system to monitor crop root health using DHT11, capacitive Soil moisture, DS18B20 and TDS sensors. Collected and processed real-time soiland environmentaldata for continuous monitoring. Integrated ThingSpeak for cloud-based data storage and visualization. Developed a web application for monitoring of live sensor data